Homework - V

Statistics III : Multivariate Data and Regression

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Question 1

Use the horseshoe crab (`crabs`) data available in the R library `asbio`. Use Y as whether a female crab has a satellite.

(a) Use a single predictor X_1 as weight and fit a logistic regression model. Report the results.

(b) For the model in part (a), what is the probability that Y is one, when X_1 is 5 kg?

(c) Plot the fitted probabilities and confidence intervals for these probabilities as a function of X_1.

(d) Use two predictors, X_1 as weight and X_2 as width to fit a logistic regression. Report the results. Which predictors are significant at 5% level?

Solution 1

The data has $n = 173$ female crabs. Let $Y_i = 1$ if the i-th female has at least one satellite (`satell` > 0) and $Y_i = 0$ otherwise; 111 crabs have $Y = 1$ and 62 have $Y = 0$. We assume $Y_i \sim \text{Bernoulli}(\pi_i)$ independently, with the logit link

$$\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 x_i$$

where x_i is the weight of the crab (in kg).

(a) Consider the following fit.

Listing 1: Logistic regression of satellite on weight

```r
library(asbio)
data(crabs)
crabs$y <- as.numeric(crabs$satell > 0)

fit1 <- glm(y ~ weight, family = binomial, data = crabs)
summary(fit1)
exp(coef(fit1))             # odds ratios
exp(confint.default(fit1)) # Wald intervals for odds ratios
```

	Estimate	Std. Error	z value	p-value
Intercept	-3.695	0.880	-4.198	2.7×10^{-5}
weight	1.815	0.377	4.819	1.4×10^{-6}

Table 1: Coefficients of the logistic regression on weight. Null deviance = 225.76 on 172 df, residual deviance = 195.74 on 171 df, AIC = 199.74.

The fitted model is

$$\boxed{\log \frac{\hat{\pi}}{1-\hat{\pi}} = -3.695 + 1.815\, x_1}$$

The slope is significant: the Wald statistic is $z = 4.819$ and the likelihood ratio statistic is $225.76 - 195.74 = 30.02$ on 1 df, giving $p = 4.3 \times 10^{-8}$. Since $e^{1.815} = 6.14$, each additional kg of weight multiplies the odds of having a satellite by about 6.14

(95% Wald interval (2.94, 12.85)). The weights have standard deviation 0.58 kg, so a more realistic increment of 0.5 kg multiplies the odds by $e^{0.5\times 1.815} \approx 2.48$. Thus, heavier females are much more likely to have a satellite. The fitted probability is 0.5 at $x_1 = 3.695/1.815 \approx 2.04$ kg.

(b) For $x_1 = 5$ the linear predictor is

$$\hat{\eta} = -3.695 + 1.815 \times 5 = 5.381$$

Thus, the estimated probability is

$$\boxed{\hat{\pi}(5) = \frac{e^{5.381}}{1 + e^{5.381}} \approx 0.9954}$$

For a confidence interval we first build one for the logit, since $\hat{\eta}$ is approximately normal. We have $\widehat{Var}(\hat{\beta}_0) = 0.7748$, $\widehat{Var}(\hat{\beta}_1) = 0.1419$ and $\widehat{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -0.3251$. This implies

$$\widehat{Var}(\hat{\eta}) = 0.7748 + 5^2(0.1419) + 2(5)(-0.3251) = 1.0715, \qquad se(\hat{\eta}) = 1.035$$

so a 95% interval for the logit is $5.381 \pm 1.96(1.035) = (3.352, 7.410)$. Transforming back with the inverse logit gives

$$\boxed{95\%\text{ CI for } \pi(5):\ (0.966,\ 0.9994)}$$

Listing 2: Probability at 5 kg

```
pr <- predict(fit1, newdata = data.frame(weight = 5), type =
    "link", se.fit = TRUE)
plogis(pr$fit)
plogis(pr$fit + c(-1, 1) * 1.96 * pr$se.fit)
```

Note that the largest weight in the data is 5.2 kg and only one crab weighs more than 4 kg. So 5 kg is at the very edge of the data and this estimate relies on very little information, even though the interval is narrow on the probability scale.

(c) We compute the fitted probability and its confidence limits on a grid of weights. The interval is computed on the logit scale, $\hat{\eta}(x) \pm 1.96\, se(\hat{\eta}(x))$, and then transformed by the inverse logit, so it always stays inside $(0, 1)$.

Listing 3: Fitted probabilities with 95% confidence band

```
grid <- data.frame(weight = seq(min(crabs$weight), max(crabs$
    weight), length.out = 200))
pr <- predict(fit1, newdata = grid, type = "link", se.fit =
    TRUE)
fitp <- plogis(pr$fit)
lo <- plogis(pr$fit - 1.96 * pr$se.fit)
up <- plogis(pr$fit + 1.96 * pr$se.fit)

png("crabs_fit.png", width = 1300, height = 840, res = 200)
plot(crabs$weight, jitter(crabs$y, amount = 0.02), pch = 16,
    col = "#96581880",
      xlab = "Weight (kg)", ylab = "P(satellite = 1)", ylim =
          c(0, 1))
polygon(c(grid$weight, rev(grid$weight)), c(lo, rev(up)), col
     = "#1b5e3c33", border = NA)
lines(grid$weight, fitp, lwd = 2, col = "#16365c")
dev.off()
```

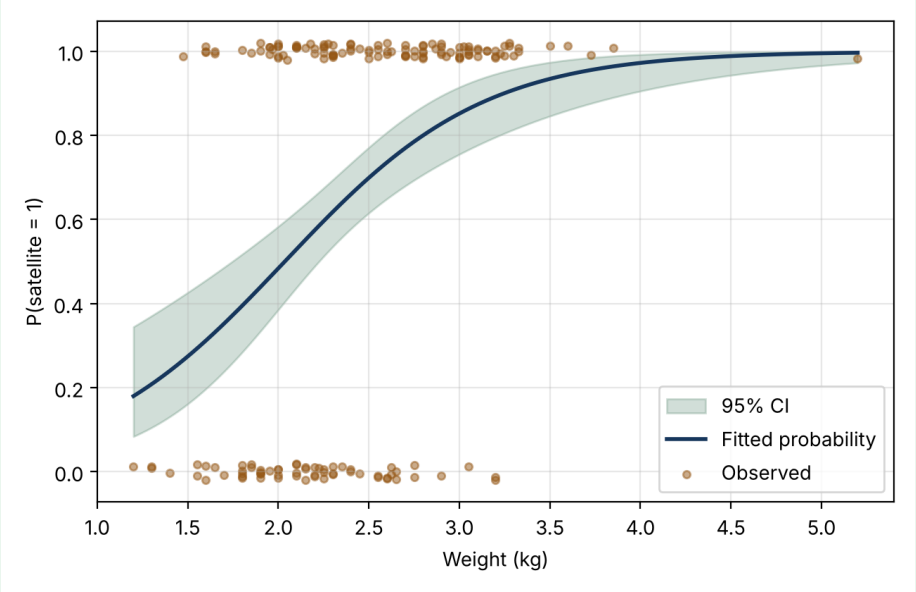


Figure 1: Fitted probability of a satellite against weight, with 95% confidence band.

The curve is the usual increasing S-shape. For example, $\hat{\pi}(1.2) = 0.18$ with interval $(0.08, 0.34)$, $\hat{\pi}(2.0) = 0.48$ with interval $(0.39, 0.58)$ and $\hat{\pi}(3.0) = 0.85$ with interval $(0.76, 0.91)$. The band is narrowest near the centre of the data (about 2.4 kg, where $se(\hat{\eta}) \approx 0.18$) and widens at both ends, where there are few crabs ($se(\hat{\eta}) = 0.45$ at 1.2 kg and 1.04 at 5 kg).

(d) Consider the model with both predictors

$$\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 \,\text{weight}_i + \beta_2 \,\text{width}_i$$

Listing 4: Logistic regression on weight and width

```r
fit2 <- glm(y ~ weight + width, family = binomial, data =
    crabs)
summary(fit2)

anova(fit1, fit2, test = "Chisq")                          #
    effect of width given weight
anova(glm(y ~ width, family = binomial, data = crabs), fit2,
    test = "Chisq")  # weight given width
cor(crabs$weight, crabs$width)
```

	Estimate	Std. Error	z value	p-value
Intercept	−9.355	3.528	−2.652	0.008
weight	0.834	0.672	1.241	0.214
width	0.307	0.182	1.686	0.092

Table 2: Coefficients of the two-predictor model. Null deviance = 225.76 on 172 df, residual deviance = 192.89 on 170 df, AIC = 198.89.

At the 5% level **neither** predictor is significant ($p = 0.214$ for weight and $p = 0.092$ for width). The same conclusion holds for the likelihood ratio tests of dropping

one predictor at a time: the deviance increases by 1.56 ($p = 0.212$) when weight is dropped and by 2.85 ($p = 0.092$) when width is dropped.

This does not mean the model is useless. The overall likelihood ratio statistic is $225.76 - 192.89 = 32.87$ on 2 df ($p = 7.3 \times 10^{-8}$), so the two predictors are jointly highly significant. The reason is that weight and width are strongly correlated ($r = 0.887$), so they carry almost the same information about Y. This inflates the standard error of the weight coefficient from 0.377 (in part (a)) to 0.672, and each predictor looks unimportant once the other is in the model. In fact, width alone is highly significant (z-test $p = 1.0 \times 10^{-6}$) and has the smallest AIC of the three models (198.45 versus 199.74 for weight alone and 198.89 for both). Thus, we do not need both predictors and one of them is enough.

Question 2

Download the German Credit data (partial) from the course website.

(a) Fit a logistic regression of GoodBad (y) on Credit amount (x). Report the estimated regression coefficient and its standard error.

(b) Plot the fitted probability of default against x along with 95% confidence intervals.

(c) Repeat step 5 by replacing the credit amount with each of the other predictors, one at a time. Find the one that is the best single predictor based on deviance criterion.

(d) Use stepwise variable selection to find the optimal model.

(e) For the selected optimal model, carry out diagnostics of the residuals.

(f) For the selected optimal model, construct a misclassification table, when the classification is 1 for fitted probability greater than 0.5 and zero otherwise.

Solution 2

The data has $n = 1000$ customers with no missing values. The response is `Good_Bad`, coded 1 for default (bad credit, 300 customers) and 0 otherwise (700 customers). The six predictors are `Duration_in_month`, `Credit_amount`, `Installment_rate`, `Present_residence_since`, `Age` and `liable_people_for_maintenance`. All of them are numeric (the last three of Installment rate, residence and liable people are small integer scales and are used as numeric). Let $\pi_i = P(Y_i = 1)$ be the probability of default.

Listing 5: Reading the data

```
library(readxl)
german <- read_excel("German_Credit_Data.xlsx")
german <- as.data.frame(na.omit(german))
str(german)
table(german$Good_Bad)
```

(a) Consider the model $\log \dfrac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 \,\text{Credit_amount}_i$.

Listing 6: Logistic regression on credit amount

```r
fit_amt <- glm(Good_Bad ~ Credit_amount, family = binomial,
    data = german)
summary(fit_amt)
```

	Estimate	Std. Error	z value	p-value
Intercept	-1.2294	0.1083	-11.348	< 0.001
Credit_amount	1.119×10^{-4}	2.355×10^{-5}	4.751	2.0×10^{-6}

Table 3: Coefficients of the logistic regression on credit amount. Null deviance = 1221.73 on 999 df, residual deviance = 1199.06 on 998 df, AIC = 1203.06.

Thus, the estimated regression coefficient and its standard error are

$$\boxed{\hat{\beta}_1 = 1.119 \times 10^{-4}, \qquad se(\hat{\beta}_1) = 2.355 \times 10^{-5}}$$

The coefficient is positive and significant ($z = 4.751$; the likelihood ratio statistic is 22.66 on 1 df). So larger loans have a higher chance of default. Since the coefficient is tiny only because the amounts are large, it is easier to read per 1000 units: the odds of default are multiplied by $e^{1000 \times 1.119 \times 10^{-4}} \approx 1.118$.

(b) We use the same approach as in Question 1: compute the interval for the logit and transform it by the inverse logit.

Listing 7: Fitted default probability with 95% confidence band

```r
grid <- data.frame(Credit_amount = seq(min(german$Credit_
    amount), max(german$Credit_amount), length.out = 300))
pr <- predict(fit_amt, newdata = grid, type = "link", se.fit
    = TRUE)
fitp <- plogis(pr$fit)
lo <- plogis(pr$fit - 1.96 * pr$se.fit)
up <- plogis(pr$fit + 1.96 * pr$se.fit)

png("german_amount.png", width = 1300, height = 840, res =
    200)
plot(grid$Credit_amount, fitp, type = "l", lwd = 2, col = "
    #16365c", ylim = c(0, 1),
      xlab = "Credit amount", ylab = "P(default)")
polygon(c(grid$Credit_amount, rev(grid$Credit_amount)), c(lo,
     rev(up)), col = "#1b5e3c33", border = NA)
lines(grid$Credit_amount, fitp, lwd = 2, col = "#16365c")
dev.off()
```

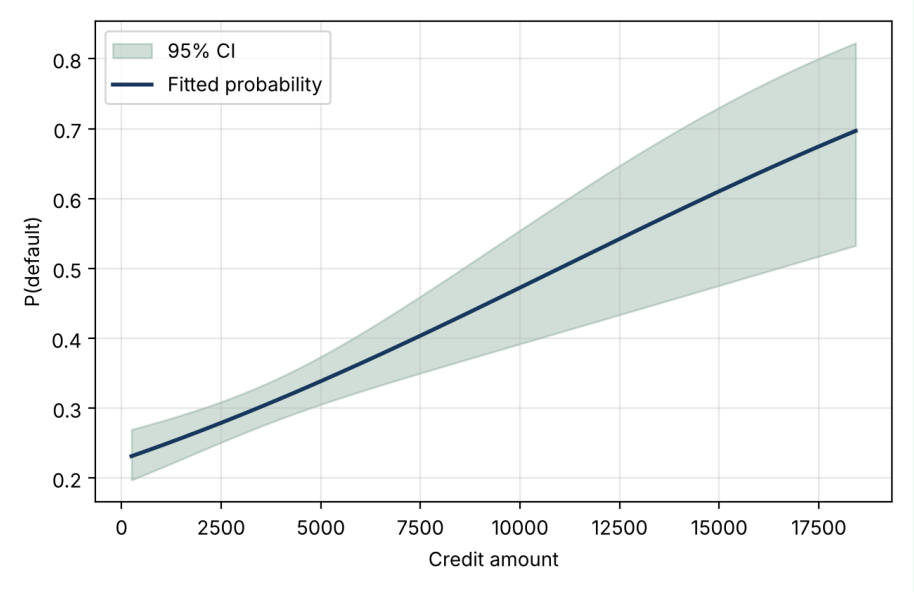


Figure 2: Fitted probability of default against credit amount, with 95% confidence band.

The fitted probability increases from 0.23 for the smallest loan (250) to 0.70 for the largest loan (18424). The band is narrow where most customers are (below about 5000) and becomes very wide for large amounts because very few loans are that large. Note that the increase in the probability is gentle, so credit amount alone cannot separate good and bad customers sharply.

(c) We interpret "step 5" as the fit in part (a). We fit the logistic regression with each predictor alone and compare the residual deviances (the null deviance is 1221.73 on 999 df). The best single predictor is the one with the smallest residual deviance, equivalently the largest drop from the null deviance. Since every model has the same number of parameters (2), the AIC gives the same ordering.

Listing 8: One-predictor models compared by deviance

```
preds <- c("Duration_in_month", "Credit_amount", "Installment_rate",
           "Present_residence_since", "Age", "liable_people_for_maintenance")
res <- t(sapply(preds, function(v) {
  f <- glm(reformulate(v, response = "Good_Bad"), family = binomial, data = german)
  c(coef = coef(f)[2], se = summary(f)$coef[2, 2], deviance = deviance(f),
    drop = f$null.deviance - deviance(f), AIC = AIC(f))
}))
res[order(res[, "deviance"]), ]
```

Predictor	Coef.	Std. Error	Resid. dev.	Drop in dev.	p-value	AIC
Duration_in_month	0.03754	0.00570	**1177.11**	44.61	2.4×10^{-11}	1181.11
Credit_amount	1.119×10^{-4}	2.36×10^{-5}	1199.06	22.66	1.9×10^{-6}	1203.06
Age	−0.01844	0.00643	1213.14	8.58	0.0034	1217.14
Installment_rate	0.14434	0.06318	1216.42	5.31	0.0212	1220.42
liable_people	−0.01823	0.19116	1221.72	0.01	0.924	1225.72
Present_residence	0.00587	0.06258	1221.72	0.01	0.925	1225.72

Table 4: One-predictor logistic regressions, sorted by residual deviance.

Thus, the best single predictor is `Duration_in_month`, with residual deviance 1177.11, i.e., a drop of 44.61 from the null deviance. Credit amount is second, and it is partly a proxy for duration since larger loans run for longer. Age and Installment rate have small but significant effects, while `Present_residence_since` and `liable_people_for_maintenance` have no effect on their own (drop in deviance ≈ 0.01 on 1 df).

(d) We use stepwise selection based on AIC, starting from the full model and allowing both addition and deletion of a variable at each step. This is the default criterion of `step()` in R.

Listing 9: Stepwise selection by AIC

```
full <- glm(Good_Bad ~ Duration_in_month + Credit_amount +
    Installment_rate +
                Present_residence_since + Age + liable_people_
                   for_maintenance,
              family = binomial, data = german)
summary(full)

best <- step(full, direction = "both")
summary(best)

# check: forward selection from the null model
null <- glm(Good_Bad ~ 1, family = binomial, data = german)
step(null, scope = formula(full), direction = "forward")
```

Step	Model (after the move)	AIC
0	all six predictors	1173.92
1	remove `liable_people` (best move; removing `Present_residence` gives 1172.22)	1172.20
2	remove `Present_residence`	1170.51
3	**stop**: no addition or deletion lowers AIC (best alternative is removing `Credit_amount`, 1172.51)	1170.51

Table 5: Path of the stepwise selection.

The selected model is

$$\boxed{\text{Good_Bad} \sim \text{Duration_in_month} + \text{Credit_amount} + \text{Installment_rate} + \text{Age}}$$

Forward selection from the null model adds Duration, Age, Installment rate and Credit amount in this order and stops at the same model, so the choice is stable. The fit is as follows.

	Estimate	Std. Error	z value	p-value	Odds ratio
Intercept	-1.5356	0.3345	-4.591	< 0.001	0.215
Duration_in_month	0.02668	0.00770	3.466	0.0005	1.027
Credit_amount	6.828×10^{-5}	3.401×10^{-5}	2.008	0.0447	1.00007
Installment_rate	0.19963	0.07229	2.762	0.0058	1.221
Age	-0.02084	0.00677	-3.079	0.0021	0.979

Table 6: Optimal model. Null deviance = 1221.73 on 999 df, residual deviance = 1160.51 on 995 df, AIC = 1170.51. The likelihood ratio statistic against the null model is 61.22 on 4 df ($p = 1.6 \times 10^{-12}$).

The deviance of the optimal model (1160.51) is almost the same as that of the full model (1159.92) even though two predictors were dropped, and the AIC improves from 1173.92 to 1170.51. Holding the other predictors fixed, we get the following interpretation.

- Each additional month of duration multiplies the odds of default by 1.027; a 12-month longer loan multiplies it by $e^{12 \times 0.02668} \approx 1.38$.
- Each additional 1000 units of credit amount multiplies the odds by $e^{0.0683} \approx 1.07$. This is only borderline significant ($p = 0.045$) because amount and duration are positively correlated.
- Each unit increase in the installment rate multiplies the odds by 1.221.
- Each additional year of age multiplies the odds by 0.979; 10 years older multiplies the odds by $e^{-10 \times 0.02084} \approx 0.81$, i.e., older customers default less.

(e) **Residual diagnostics.** For the optimal model we examine the deviance residuals d_i, the Pearson residuals, the leverages h_i and Cook's distances.

Listing 10: Diagnostics for the optimal model

```
library(arm)
library(ResourceSelection)
dres <- residuals(best, type = "deviance")
pres <- residuals(best, type = "pearson")
sres <- rstandard(best)                    # standardized
    deviance residuals
h    <- hatvalues(best)
cd   <- cooks.distance(best)
eta  <- predict(best, type = "link")

png("german_diag.png", width = 2200, height = 1275, res =
    170)
par(mfrow = c(2, 3))
plot(eta, dres, xlab = "linear predictor", ylab = "deviance
    residual",
     main = "Deviance residuals vs linear predictor"); abline
        (h = 0)
plot(eta, pres, xlab = "linear predictor", ylab = "Pearson
    residual",
     main = "Pearson residuals vs linear predictor"); abline(
        h = 0)
binnedplot(fitted(best), residuals(best, type = "response"),
    nclass = 40)
qqnorm(sres); qqline(sres)
plot(h, sres, xlab = "leverage", ylab = "std. deviance
    residual"); abline(v = 3 * 5 / 1000, lty = 2)
```

```r
plot(cd, type = "h", ylab = "Cook's distance", main = "Cook's
    distance")
dev.off()

max(h); sum(h > 3 * 5 / 1000)
sum(abs(sres) > 2); max(abs(sres))
max(cd); which.max(cd)
hoslem.test(german$Good_Bad, fitted(best), g = 10)
```

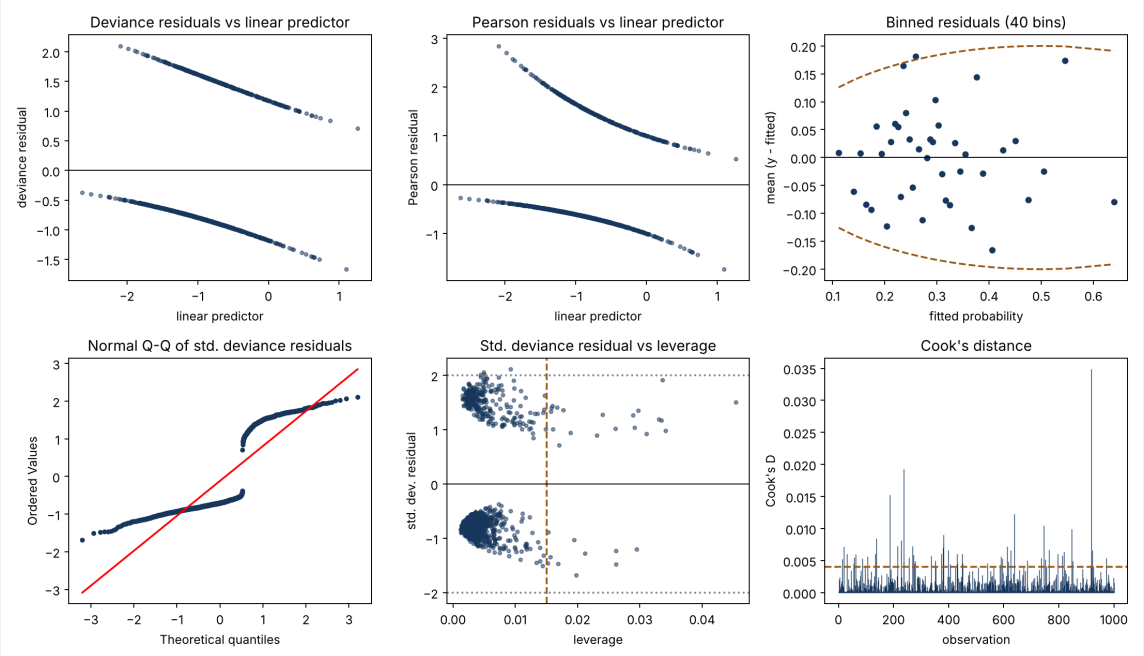


Figure 3: Diagnostic plots of residuals for the optimal model.

Since the response is binary, the raw residuals can only take one of two values for each fitted value, so they fall on two smooth curves (one for $y_i = 0$ and one for $y_i = 1$) in the residual plots. Also the Q-Q plot has the step shape that is typical for binary responses. Thus, the usual "no pattern" and "straight line" checks are not meaningful here and we should not expect normality of the residuals. We look at the following instead.

- **Binned residuals.** After grouping the observations into 40 bins of equal size by fitted probability, the average of $y_i - \hat{\pi}_i$ lies inside the $\pm 2\,se$ limits for 39 of the 40 bins, with no trend, so the fitted probabilities are well calibrated across the range (0.07 to 0.78).
- **Goodness of fit.** The residual deviance (1160.5 on 995 df) cannot be used for a χ^2 test with ungrouped binary data. The Hosmer–Lemeshow test with 10 groups gives a statistic of about 6.8 on 8 df ($p \approx 0.56$), so there is no evidence of lack of fit. (The value from `hoslem.test` may differ slightly in the second decimal because of the grouping rule.)
- **Outliers.** The deviance residuals lie between -1.66 and 2.10. Only 4 of the 1000 standardized deviance residuals exceed 2 in absolute value, the largest being 2.11, and none exceeds 3.

- **Leverage.** The maximum leverage is 0.045 against an average of $p/n = 0.005$, and 30 observations exceed $3p/n = 0.015$. These are customers with unusually large amounts or long durations, but all of them have moderate residuals (see the leverage plot).
- **Influence.** The maximum Cook's distance is 0.035 (observation 918: a 6-month loan of 14896 to a 68-year-old customer who defaulted although the fitted probability was only 0.17). This is far below 0.5, so no observation is influential.

We also check that the logit is linear in the continuous predictors by plotting the empirical logit of the default rate within quantile groups of each predictor.

Listing 11: Empirical logit plots

```
png("german_logit.png", width = 2200, height = 650, res =
    170)
par(mfrow = c(1, 3))
for (v in c("Duration_in_month", "Credit_amount", "Age")) {
  g <- cut(german[[v]], quantile(german[[v]], 0:10 / 10),
      include.lowest = TRUE)
  p <- (tapply(german$Good_Bad, g, sum) + 0.5) / (tapply(
      german$Good_Bad, g, length) + 1)
  plot(tapply(german[[v]], g, mean), log(p / (1 - p)), type =
      "b", pch = 16,
       xlab = v, ylab = "empirical logit")
}
dev.off()
```

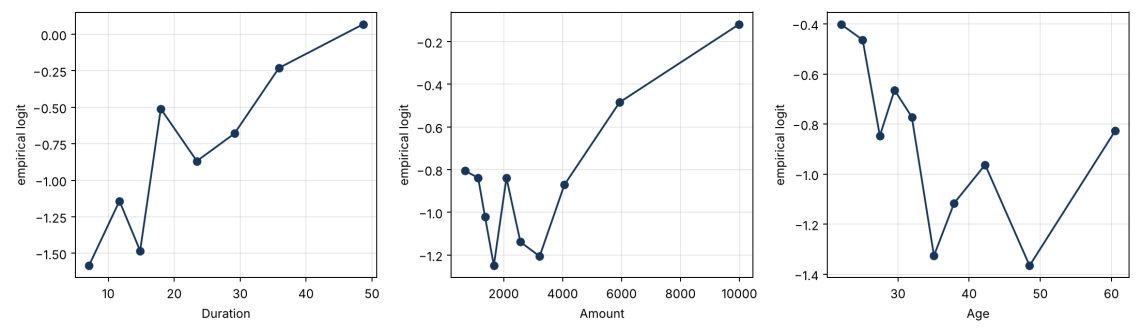


Figure 4: Empirical logit of the default rate against each continuous predictor.

The empirical logits are noisy, since each group has only about 100 observations, but they increase with duration and decrease with age, as the fitted coefficients say. For credit amount the logit is flat up to about 3000 and then rises, which is a mild departure from linearity (a transformation such as log(amount) or a spline could be tried). Overall, the residual diagnostics show no serious violation of the model.

(f) **Misclassification table.** We classify a customer as a defaulter ($\hat{y}_i = 1$) if $\hat{\pi}_i > 0.5$ and as a non-defaulter otherwise.

Listing 12: Misclassification table at cut-off 0.5

```
pred <- as.numeric(fitted(best) > 0.5)
tab <- table(Actual = german$Good_Bad, Predicted = pred)
```

```
tab
sum(diag(tab)) / sum(tab)    # accuracy
tab[2, 2] / sum(tab[2, ])    # sensitivity
tab[1, 1] / sum(tab[1, ])    # specificity
```

	Predicted 0	**Predicted 1**	**Total**
Actual 0	674	26	700
Actual 1	260	40	300
Total	934	66	1000

Table 7: Misclassification table of the optimal model (cut-off 0.5).

This gives

$$\boxed{\text{misclassification rate} = \frac{26 + 260}{1000} = 0.286}$$

i.e., an accuracy of 71.4%. The specificity is $674/700 = 96.3\%$ but the sensitivity is only $40/300 = 13.3\%$. Thus, the model finds very few of the actual defaulters. The reason is that only 30% of the customers default and the largest fitted probability is 0.78, so only 66 customers cross the cut-off of 0.5. Note that the trivial rule "predict 0 for everyone" already has accuracy 70%, so the model improves on it only slightly at this cut-off. If catching defaulters matters more than avoiding false alarms, the cut-off should be lowered towards the default rate of 0.3, at the cost of lower specificity.

Question 3

Consider the data used in class on distribution of opinions about marijuana possession before and after television series has aired. The distribution before was Decriminalization (7%), Legalization (18%), Existing law (65%), No Opinion (10%). After the series aired, 500 individuals were randomly selected and interviewed. Their opinions were recorded as Decriminalization (39), Legalization (99), Existing law (336), No Opinion (26).

(a) Set this up as a hypothesis testing problem, specifying the null and alternative hypotheses, parameter, data and test statistic.

(b) Use the result that the distribution of the test statistic is asymptotically chi-squared, to obtain the p-value, stating the appropriate degrees of freedom.

(c) The function `multinom_test` in the R-library `rstatix` carries out an exact test in this situation. Find the p-value using this method.

(d) The exact test takes a long time and the asymptotic test is approximate. Another possibility is to generate random vectors from the multinomial distribution under the null hypothesis using the function `rmultinom`, computing the test statistic of this random vector and simulating the distribution of the test statistic under the null hypothesis. Use a sample of size 10^5 to obtain the p-value using this method.

Solution 3

(a) **Set-up.** The parameter is the vector of population proportions of the four opinions after the series, $p = (p_1, p_2, p_3, p_4)$ with $\sum_j p_j = 1$ (Decriminalization, Legalization, Existing law, No Opinion). The data are the counts $(O_1, O_2, O_3, O_4) = (39, 99, 336, 26)$ with $n = 500$, and we assume

$$(O_1, O_2, O_3, O_4) \sim \text{Multinomial}(500;\, p_1, p_2, p_3, p_4)$$

The question is whether the distribution of opinions changed after the series, i.e., whether p is still equal to the earlier distribution $p^0 = (0.07,\, 0.18,\, 0.65,\, 0.10)$. Thus, we test

$$H_0 : p = p^0 = (0.07, 0.18, 0.65, 0.10) \quad \text{against} \quad H_1 : p \neq p^0$$

(i.e., $p_j \neq p_j^0$ for at least one j). The expected counts under H_0 are $E_j = np_j^0$ and the test statistic is Pearson's chi-squared statistic

$$\boxed{X^2 = \sum_{j=1}^{4} \frac{(O_j - E_j)^2}{E_j}}$$

Large values of X^2 are evidence against H_0, so the test rejects for large X^2.

(b) The expected counts are $E = (35,\ 90,\ 325,\ 50)$, all larger than 5, so the asymptotic approximation is reasonable. We have

$$X^2 = \frac{(39-35)^2}{35} + \frac{(99-90)^2}{90} + \frac{(336-325)^2}{325} + \frac{(26-50)^2}{50}$$

$$= 0.457 + 0.900 + 0.372 + 11.520 = 13.249$$

Under H_0, X^2 is asymptotically χ^2_{k-1} with $k - 1 = 4 - 1 = 3$ degrees of freedom, as no parameter is estimated from the data. Thus, the p-value is

$$\boxed{p = P(\chi^2_3 \geq 13.249) \approx 0.0041}$$

The 5% critical value is $\chi^2_{3,\,0.05} = 7.815 < 13.249$, so we reject H_0 at the 5% level (and also at 1%, as $0.0041 < 0.01$). The "No Opinion" category contributes 11.52 of the total 13.25: only 26 of the 500 (5.2%) have no opinion, against 10% before. Thus, the change is mainly that fewer people are undecided after the series; the other three categories are close to their earlier proportions (7.8%, 19.8% and 67.2%).

Listing 13: Asymptotic chi-squared test

```r
obs <- c(39, 99, 336, 26)
p0  <- c(0.07, 0.18, 0.65, 0.10)
E   <- 500 * p0
X2  <- sum((obs - E)^2 / E)
X2
pchisq(X2, df = 3, lower.tail = FALSE)
chisq.test(obs, p = p0)      # same result
```

(c) The exact test computes the probability, under H_0, of every possible outcome $(x_1, \ldots, x_4)$ with $\sum_j x_j = 500$. The p-value is the total probability of all outcomes that are no more likely than the observed one,

$$p_{\text{exact}} = \sum_{x:\ P_{H_0}(x) \leq P_{H_0}(O)} P_{H_0}(x), \qquad P_{H_0}(x) = \frac{500!}{x_1!\, x_2!\, x_3!\, x_4!} \prod_{j=1}^{4} (p_j^0)^{x_j}$$

There are $\binom{503}{3} = 21{,}084{,}251$ possible outcomes, which is why this test is slow.

Listing 14: Exact multinomial test

```r
library(rstatix)
multinom_test(obs, p = p0)
```

Enumerating all 21,084,251 outcomes gives

$$\boxed{p_{\text{exact}} \approx 0.0016}$$

This is smaller than the asymptotic p-value (0.0041), but the conclusion is the same: we reject H_0. The two numbers differ partly because the exact test orders the outcomes by their probability, while X^2 orders them by their distance from E. If we use X^2 to order the outcomes in the exact test, i.e., sum the probabilities of all outcomes with $X^2 \geq 13.249$, we get $p = 0.0043$, which is very close to the asymptotic value.

(d) Under H_0 we generate $B = 10^5$ multinomial vectors of size 500 with probabilities p^0, compute X^2 for each and estimate the p-value by the proportion of simulated statistics that are at least as large as the observed one,

$$\hat{p} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{X_b^2 \geq X_{\text{obs}}^2\}$$

Listing 15: Monte Carlo simulation of the null distribution

```r
set.seed(2026)
B   <- 10^5
sim <- rmultinom(B, size = 500, prob = p0)       # 4 x B matrix
X2_sim <- colSums((sim - E)^2 / E)
mean(X2_sim >= X2)                               # simulated p-value
hist(X2_sim, breaks = 60, freq = FALSE, main = "Null distribution of X^2")
curve(dchisq(x, df = 3), add = TRUE, lwd = 2)
```

The simulation gives

$$\boxed{\hat{p} \approx 0.0043}$$

Its standard error is $\sqrt{\hat{p}(1-\hat{p})/B} \approx 0.0002$, so the value changes only in the fourth decimal with the random seed. The histogram of the simulated statistics follows the χ^2_3 density closely, which supports the asymptotic approximation.

Method	**p-value**
Asymptotic χ^2_3	0.0041
Exact test (outcomes ordered by probability)	0.0016
Exact test (outcomes ordered by X^2)	0.0043
Simulation, $B = 10^5$	≈ 0.0043

Table 8: Comparison of the four p-values.

All the methods give a p-value below 0.01. Thus, we reject H_0: there is strong evidence that the distribution of opinions changed after the series. The simulation is very close to the asymptotic and the chi-squared based exact value, and it takes only seconds compared to the full enumeration.

Question 4

A social scientist wants to determine whether the marital status (divorced or not divorced) of U.S. men is independent of their religious affiliation (A–D) or lack thereof. A sample of 500 U.S. men is surveyed, and the results are tabulated as shown in the following table.

	A	B	C	D	None	Totals
Divorced	39	19	12	28	18	116
Married, never divorced	172	61	44	70	37	384
Totals	211	80	56	98	55	500

Test to see whether there is sufficient evidence to indicate that the marital status of men who have been or are currently married is dependent on religious affiliation. Take $\alpha = 0.01$.

Solution 4

We have a 2×5 contingency table with $n = 500$. Let O_{ij} be the observed count in row i (marital status) and column j (religious affiliation), with row totals r_i and column totals c_j. We test

H_0 : marital status and religious affiliation are independent against H_1 : they are dependent

Under H_0 the cell probabilities factor as $p_{ij} = p_{i\cdot}\, p_{\cdot j}$, so the expected counts are estimated by

$$\hat{E}_{ij} = \frac{r_i c_j}{n}$$

and the test statistic is

$$\boxed{X^2 = \sum_{i=1}^{2} \sum_{j=1}^{5} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}}$$

which is asymptotically $\chi^2_{(2-1)(5-1)} = \chi^2_4$ under H_0. For example, $\hat{E}_{11} = \dfrac{116 \times 211}{500} = 48.952$. The expected counts and the contributions to X^2 are as follows.

	A	**B**	**C**	**D**	**None**
Expected counts $\hat{E}_{ij}$					
Divorced	48.952	18.560	12.992	22.736	12.760
Married, never divorced	162.048	61.440	43.008	75.264	42.240
Contributions $(O_{ij} - \hat{E}_{ij})^2/\hat{E}_{ij}$					
Divorced	2.023	0.010	0.076	1.219	2.152
Married, never divorced	0.611	0.003	0.023	0.368	0.650

Table 9: Expected counts and contributions to the chi-squared statistic.

All the expected counts are at least $12.76 > 5$, so the chi-squared approximation is valid. Adding the contributions gives

$$X^2 = 2.023 + 0.010 + 0.076 + 1.219 + 2.152 + 0.611 + 0.003 + 0.023 + 0.368 + 0.650 = 7.135$$

The degrees of freedom are $(2-1)(5-1) = 4$. Thus, the critical value at $\alpha = 0.01$ is $\chi^2_{4,\,0.01} = 13.277$ and the p-value is

$$\boxed{p = P(\chi^2_4 \geq 7.135) \approx 0.129}$$

Listing 16: Chi-squared test of independence

```r
tab <- matrix(c(39, 19, 12, 28, 18,
                172, 61, 44, 70, 37), nrow = 2, byrow = TRUE,
              dimnames = list(c("Divorced", "Married"), c("A", "B
                ", "C", "D", "None")))
test <- chisq.test(tab)
test
test$expected
qchisq(0.99, df = 4)
```

Conclusion. Since $X^2 = 7.135 < 13.277$ (equivalently $p = 0.129 > 0.01$), we do not reject H_0. At the 1% level there is not sufficient evidence that the marital status of men who have been or are currently married depends on their religious affiliation. The same conclusion holds even at the 5% level, where the critical value is 9.488. The sample proportions of divorced men do vary across the groups (18.5% for A, 23.8% for B, 21.4% for C, 28.6% for D and 32.7% for None, against 23.2% overall), but a difference of this size is quite plausible by chance with the group sizes here (the largest contributions come from group A and the "None" group, and even they are only about 2 each).